Phase III: Standalone Performance Analysis Report

Overview of Evaluation Methodology

The performance analysis evaluates the predictive capabilities of two tree-based regression models—Random Forest and XGBoost—for short-term solar power forecasting. The evaluation utilized the DKASC Alice Springs dataset, enforcing a strict chronological train-validation split (80/20) to prevent data leakage and preserve the time-series integrity of the environmental data. The models were trained on 153,900 observations and validated against 38,475 unseen observations.

Error Metrics and Theoretical Interpretation

The models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R-squared).

Model	R-squared (R^2)	RMSE (kW)	MAE (kW)
Random Forest	**0.9651**	**1.5450**	**0.7838**
XGBoost	0.9627	1.5968	0.8090

Both algorithms proved highly effective, explaining over 96% of the variance in solar power generation. However, Random Forest consistently produced lower error margins across both RMSE and MAE.

The Random Forest architecture utilizes bagging (Bootstrap Aggregating), which builds numerous independent decision trees in parallel and averages their predictions. This methodology proved highly resilient to the inherent noise and high variance found in the non-linear meteorological data. Conversely, XGBoost utilizes a sequential boosting technique, optimizing each subsequent tree to correct the residuals of the previous ones. While powerful, the boosting process appears slightly more sensitive to outlier fluctuations in the solar irradiance data within this specific dataset configuration, resulting in marginally higher error metrics compared to the parallelized stability of the Random Forest.

Visual Validation

The quantitative metrics are supported by visual validation (refer to the Actual vs. Predicted Power Generation chart). The predictive curves of both models accurately map the diurnal solar generation cycles, demonstrating high fidelity during peak production hours and correctly regressing to zero during nighttime periods without predictive latency.

Conclusion

While both models successfully satisfy the Phase III implementation requirements for robust predictive modeling, the Random Forest model is identified as the superior algorithm for this specific forecasting task due to its lower error rates and robust generalization capabilities.